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ENGLISH MEDIUM

PARMAR SSC

LITTLE BOY

GK MCQ BOOK

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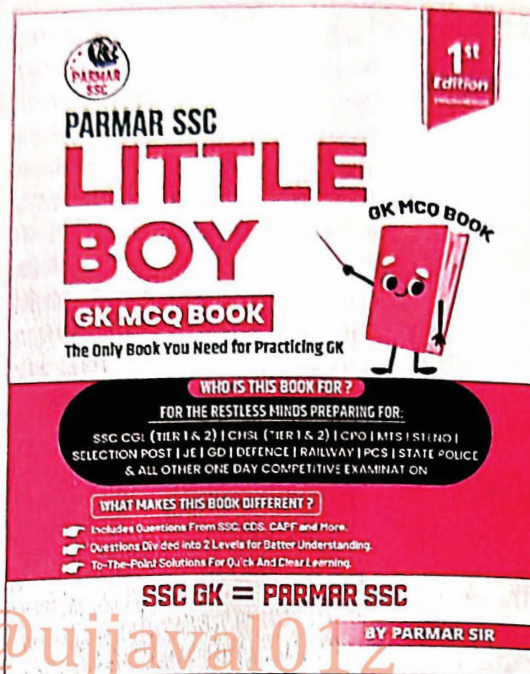
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BY PARMAR SIR

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- | | |
|------------------|---|
| ➤ AUTHORED BY | - PARMAR SIR |
| ➤ STRATEGIZED BY | - SANJOG SAINI |
| ➤ CHIEF EDITOR | - RITESH SAROJ |
| ➤ PROOFREADERS | - RITESH SAROJ
SHIVANI BHARTI
ADITI JASSI |
| ➤ COVER PAGE | - HARSHIT RAJPUT |

Price: ₹ 599/-

PUBLISHER



DISTRIBUTOR



For Any Query : ☎ 9250604879

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Preface

In the ever-evolving landscape of competitive exams, precision and practice defines progress. With that vision, I present **LITTLE BOY**—a carefully curated collection of MCQs designed not merely to test, but to train. This book is the result of countless hours of selection, verification, and fine-tuning—guided by purpose and powered by the collective strength of those who believe in consistent excellence. I owe my deepest gratitude to my family and the entire **PARMARian** community, whose steadfast encouragement turned this project into a mission.

Each question in this book carries a purpose—to test, to teach, and to transform. These are not random MCQs scattered on pages; they are handpicked, battle-tested, and refined to sharpen your instincts. It's not about just choosing an option—it's about choosing confidence.

This isn't a book that promises shortcuts. It's a mirror of reality. It challenges you to show up every day, face your weaknesses, and grow beyond them. And through that grind, **LITTLE BOY** will be your loyal comrade.

So Go Ahead—Flip The Page, Fire Up Your Brain, And Fight Smart.

THANKOO;)

PARMAR SIR

LITTLE STUDENT MANUAL

TURNING PAGES INTO PROGRESS

Let's be honest-GK/GS isn't about reading endless pages. It's about practicing the right questions, understanding concepts quickly, and being exam-ready. That's exactly why LITTLE was created.

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5. REMEMBER - IT'S ABOUT SMART WORK, NOT HARD WORK!

- ❖ This book isn't to decorate your shelf. It's designed to be used daily, even if just for 15 minutes.

The more you interact with it, the closer your get to clearing that exam. So, open a Topic, Attempt, Scan, Learn and Keep moving forward. With LITTLE, you're not just practicing-you're preparing the way toppers do. "Because in competitive exams, it's not about knowing everything-it's about knowing exactly what matters."

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Let every page that follows bring you a step closer to understanding and action.

”



LEVEL I

1. In 1905, who concluded with Thomas Chamberlain that the planets of the Solar System originated from an encounter between the Sun and another star? [SSC CPO 27/06/2024]
A) Forest Moulton B) Inge Lehmann
C) Charles Lyell D) James Hutton
2. Which dwarf planet orbits the sun in 310 years at a distance between 38.5 to 53 AU? [SSC CHSL 03/07/2024]
A) Ceres B) Eris
C) Makemake D) Pluto
3. Which of the following statement is correct? [SSC CHSL 04/07/2024]
A) The density of Saturn is similar to that of Earth.
B) The Size of Earth and Venus is almost similar.
C) The size of Jupiter and Mars is similar.
D) Mars is the warmest planet.
4. Which of the following does NOT have rings? [SSC CHSL 11/07/2024]
A) Venus B) Uranus
C) Saturn D) Jupiter
5. ____ oblong orbit overlaps the orbit of Neptune. [SSC CHSL 10/07/2024]
A) Venus' B) Mars'
C) Pluto's D) Jupiter's
6. Which of the following rocky planets is the fastest planet in our solar system-travelling through space at 29 miles (47 Kilometers) per second? [SSC CHSL 05/07/2024]
A) Earth B) Mercury
C) Venus D) Mars
7. The length of a day on ____ is approximately 16 hours. [SSC CHSL 05/07/2024]
A) Mars B) Mercury
C) Venus D) Neptune
8. Which of the following planets' atmosphere is made up of thick white and yellowish clouds of sulphuric acid? [SSC CHSL 03/10/2020]
A) Jupiter B) Neptune
C) Venus D) Mars
9. Which of the following statements about planet Saturn is true? [SSC CHSL 04/07/2024]
A) Saturn is the fourth planet from the Sun and the largest planet in our Solar system.
B) Satellites like Enceladus and Titan of Saturn are home to internal oceans, could possibly support life.
C) Saturn is a massive ball made up mostly of nitrogen.
D) Saturn's environment is conducive to life.
10. Which planet is the second largest planet in our solar system and also has the biggest and brightest rings around it? [SSC CHSL 10/07/2024]
A) Jupiter B) Earth
C) Uranus D) Saturn
11. Which is the most widely accepted model to explain the formation and evolution of the solar system? [SSC CPO 03/10/2023]
A) Cloud hypothesis B) Gas hypothesis
C) Nebular hypothesis D) Solar hypothesis
12. Terrestrial planets are composed of _____. [SSC CPO 03/10/2023]
A) Metals and air B) Rocks and metals
C) Metals and gas D) Rocks and gas
13. Which trend is found about the distance between the galaxies? [SSC CPO 04/10/2023]
A) Unpredictable B) Decreasing
C) Constant D) Increasing
14. Which scientist thought of the concept of the steady state of the universe? [SSC CGL 14/07/2023]
A) Harold Jeffrey B) Edwin Hubble
C) Fred Hoyle D) Pierre-Laplace
15. Which among the following is another word for universe? [SSC CHSL 10/03/2023]
A) Big Crunch B) Astronomy
C) Cosmos D) Supernova
16. Which of the following is a part of celestial objects? [SSC CPO 09/11/2022]
I. Stars II. Planets
III. Moon
A) Only I and III B) Only I and II
C) I, II and III D) Only II and III
17. The bodies glowing in the night sky are known as _____. [SSC CGL 03/03/2020]
A) Celestial bodies B) Asteroids
C) Planetarium D) Meteorite
18. Cosmic snowballs of frozen gases, rock and dust that orbit the Sun are known as _____. [SSC MTS 16/05/2023]
A) Meteors B) Star
C) Meteorites D) Comets
19. Which of the following constellations is also called the hunter? [SSC CHSL 10/03/2023]
A) Cassiopela B) Leo Major
C) Ursa Major D) Orion
20. Which among the following planets is also known as "Morning Star"? [SSC MTS 10/05/2023]
A) Jupiter B) Neptune
C) Venus D) Earth
21. Which among the following is the coldest planet in the solar system? [SSC MTS 16/05/2023]
A) Mars B) Neptune
C) Saturn D) Uranus
22. The Kuiper Belt is a ring of icy rocks & dust bodies just outside the orbit of _____. [SSC MTS 17/05/2023]
A) Saturn B) Venus
C) Jupiter D) Neptune
23. The asteroid belt lies between which two planets? [SSC CHSL 21/03/2023]
A) Jupiter and Saturn B) Mars and Jupiter
C) Saturn and Uranus D) Mercury and Venus

24. In approximately how much time does Mars complete one spin on its axis?
A) 16 hours B) 8 hours [SSC CHSL 15/03/2023]
C) 1 day 18 hours D) 1 day
25. On which planet has a rotating oval of clouds, twice as wide as Earth, called the 'Great Red Spot' been observed for more than 300 years?
A) Neptune B) Mercury [SSC CHSL 08/08/2023]
C) Jupiter D) Venus
26. The distance of the Sun from the Earth is about _____ light minutes.
A) 6 B) 4.3 [SSC CHSL 20/03/2023]
C) 12 D) 8
27. Who was the first to accurately describe the rings of Saturn as a disc around the planet in 1655?
A) Hideki Yukawa B) Galileo Galilei [SSC CGL 06/12/2022]
C) Christiaan Huygens D) Giovanni Cassini
28. From the given alternatives, identify the dwarf planet.
A) Pluto B) Mercury [SSC CHSL 30/06/2023]
C) Neptune D) Mars
29. The _____ radiation belts are giant swaths of magnetically trapped highly energetic charged particles that surround Earth.
A) Van Allen B) Aurora [SSC CGL 13/04/2022]
C) Kuiper D) Chinook
30. Which of the following is the windiest planet in the solar system?
A) Neptune B) Uranus [SSC MTS 14/10/2021]
C) Saturn D) Mars
31. Lucifer is another name for the planet _____.
A) Jupiter B) Mars [SSC CGL 03/03/2020]
C) Venus D) Saturn
32. Minal wrote an essay on the largest moon of Saturn. Which among the following moons is it?
A) Atlas B) Titan [SSC MTS 19/06/2023]
C) Carme D) Europa
33. The moon moves around the earth in about _____ days.
A) 21 B) 23 [SSC MTS 14/06/2023]
C) 27 D) 25
34. In early 1610, who discovered with his newly invented telescope that Jupiter has four moons?
A) Simon Marius B) Tycho Brahe [SSC CGL 06/12/2022]
C) Galileo Galilei D) Johannes Kepler
35. Which of the following celestial bodies has a natural satellite named 'Charon'?
A) Haumea B) Pluto [SSC MTS 05/10/2021]
C) Mars D) Saturn
36. What is the radius of the moon?
A) 1.78×10^6 m B) 1.79×10^6 m [SSC CHSL 19/03/2020]
C) 1.74×10^5 m D) 1.74×10^6 m
37. Planetary scientists call the gaseous envelope around the Moon as the _____.
A) lunar exosphere B) lunar stratosphere [SSC CGL 04/03/2020]
C) lunar thermosphere D) lunar endosphere
38. In which of the following periods is the meteor shower named Quadrantids generally visible from Earth?
A) August/September B) October/November [SSC MTS 20/10/2021]
C) May/June D) December/January
39. In which of the following months may the meteor shower named Lyrids be seen from Earth?
A) April B) February [SSC MTS 20/10/2021]
C) June D) August
40. Each orbit of the International Space Station (ISS) takes _____ minutes.
A) 80 - 82 B) 85 - 87 [SSC CHSL 13/03/2023]
C) 90 - 93 D) 83 - 86
41. The moon completes _____ rotation on its axis as it completes one revolution around the Earth.
A) 3 B) 2 [SSC CPO 09/11/2022]
C) 1 D) 4
42. A distinctive cross-shaped constellation best seen in the northern hemisphere during the summer and fall months around September is _____.
A) Cygnus B) Pegasus [SSC CHSL 02/07/2024]
C) Ursa Major D) Cassiopeia

LEVEL II

1. Which one among the following planets has the largest number of known satellites?
A) Mars B) Neptune [CDS II 2022]
C) Jupiter D) Saturn
2. Which of the following groups of planets is termed as 'gas planets' as they are composed primarily of lighter ices, liquids and gases?
A) Mars, Jupiter, Neptune, Uranus
B) Jupiter, Uranus, Neptune, Saturn
C) Saturn, Mars, Jupiter, Neptune
D) Neptune, Saturn, Mars, Uranus
3. Consider the following statements about some planets in the solar system:
1. Mercury has no atmosphere.
2. Venus has two moons.
3. There is no land on Jupiter.
4. Rings of Saturn are composed of lumps of ice and dust.

Which of the statements given above are correct?

- A) 1 and 2 only B) 3 and 4 only [CDS II 2023]
C) 1, 2 and 3 D) 1, 3 and 4
4. In Sun-earth system, the Sun, the Earth and Lagrange point L4 form:
A) An isosceles triangle B) An equilateral triangle [CDS II 2024]
C) A straight line D) A scalene triangle
5. In Sun-Earth system, the distance between Lagrange points L2 and L3 is about _____
A) 15 lakh kilometres B) 30 lakh kilometres [CAPF 2024]
C) 16 crore kilometres D) 32 crore kilometres
6. Which one of the following statements about the solstices, an event that occurs when the Sun appears to reach most northerly or southerly, is correct?
[CAPF 2024]

- A) The winter solstice takes place on June 21 in both the northern and southern hemispheres.
 B) The winter solstice takes place on December 21 in both the northern and southern hemispheres.
 C) The summer solstice occurs in northern hemisphere on June 21 and on December 21 in southern hemisphere.
 D) The summer solstice occurs in northern hemisphere on December 21 and on June 21 in southern hemisphere.
7. Which one of the following statements with reference to the **Nebular Hypothesis** for the origin of the Solar System is correct? [CAPF 2024]
 A) The initiation of the nebular hypothesis can be considered as a diffuse, roughly spherical, slowly rotating expanding nebula.
 B) As a result of rotation and expansion, a disk is formed with matter concentrated in the centre.
 C) Expansion continues with the formation of the proto-Sun and the rings of material are left behind.
 D) The material in the rings get condensed and form the planets that revolve in orbits around the Sun.
8. According to the Big Bang theory of the origin of the Universe, the **age of the Universe** is estimated at about: [CISF AC 2023]
 A) 0 - 3 million years B) 4 - 6 billion years
 C) 7 - 10 billion years D) 13 - 18 billion years
9. Which one of the following planets has the highest density? [NDA I 2021]
 A) Mercury B) Venus
 C) Jupiter D) Earth
10. Which one from among the following planets is largest in size? [NDA I 2023]
 A) Earth B) Venus
 C) Mars D) Mercury
11. Which one of the following astronomers proved that the Earth and other planets revolve around the Sun? [NDA I 2023]
 A) Copernicus B) Kepler
 C) Galileo D) Newton
12. Which one of the following is the correct sequence of arrangement of the given planets in descending order of their density (in gm/cm³)? [NDA I 2023]
 A) Earth > Jupiter > Venus > Saturn
 B) Jupiter > Earth > Saturn > Venus
 C) Earth > Venus > Jupiter > Saturn
 D) Earth > Venus > Saturn > Jupiter
13. Which of the following are called 'terrestrial planets'? [GEOSCIENTIST 2022]
 A) Earth, Mars, Jupiter, Mercury
 B) Jupiter, Mercury, Saturn, Earth
 C) Earth, Mercury, Mars, Venus
 D) Mercury, Mars, Earth, Saturn

@ujjaval012

17. (A) Celestial bodies

Planetarium - a building with curved ceiling built for educational and entertaining purpose

Meteorite - meteor hitting the Earth surface

Meteoroid - any solid debris of asteroid, comet or other celestial objects that floats through interplanetary space

Meteor (shooting/falling star) - streak of light that appears when a meteoroid enters the atmosphere (mesosphere)

See solution of question 16 (level 1)

18. (D) Comets

See solution of question 16 (level 1)

Stars - luminous bodies, their colour depends on temperature

Group of stars - Constellation (Largest - Hydra, Ursa Major - Sapt Rishi).

Do you know that Rigel is the **brightest star** in Orion constellation while **overall Sirius** (dog star) is the brightest star?

Closest star to Earth - Sun (150 mn km), **2nd closest** - Proxima Centuri

19. (D) Orion

Cassiopeia - W shaped

Cygnus - cross shaped

January evenings - hourglass shaped

Asterism is the pattern of stars smaller than constellation

20. (C) Venus

See solution of question 8

Jupiter - winter planet, largest, 3rd brightest after Venus and Moon, shortest rotation (10 hrs), **95 moons** (Galilean moons - Io, Europa, Ganymede {largest}, Calisto), has unclear rings

21. (D) Uranus

Uranus - Green planet (due to CH_4), rolling/lopsided planet (tilted 98°), discovered by William Herschel (1781)

Mars - Red planet (iron oxide), **2 moons** - Phobos and Deimos, largest volcano and tallest mountain - Olympus Mons

22. (D) Neptune

See solution of question 5 (level 1)

23. (B) Mars and Jupiter

24. (D) 1 day

Martian day (sol) is of about **24.6 hours**.

Rotation time of other planets is - Venus - 243 days (slowest), Earth - 24 hours, Jupiter - 10 hours (fastest)

25. (C) Jupiter

Great Red Spot is a storm system on Jupiter.

More info about Jupiter - see solution of question 20

Sunspots are temporary phenomena on the photosphere of the Sun appearing as dark spots as they are $500-1500^\circ\text{C}$ cooler than the surrounding chromosphere.

26. (D) 8

Sun - 99% mass of the solar system, 109 time of Earth, composed of Hydrogen (73%) and Helium (25%), outer layer - Corona, visible part - Photosphere

Light year (9.46×10^{12} km) is the distance travelled by light in one year. 1 Parsec = 3.26 light year.

Do you know that Aditya L1 is the first solar mission of India?

27. (C) Christian Huygens

See solution of question 9 (level 1)

28. (A) Pluto

See solution of questions 5 and 2 (level 1)

29. (A) Van Allen

Van Allen belts are two **radiation belts** of charged particles located in Earth's magnetic field.

Location - Inner belt - 3,000 km above Earth; outer belt - 15,000-20,000 km above Earth.

These belts are dangerous for humans and spacecrafts and the energetic electrons in them are called "**killer**" electrons as they can damage spacecrafts.

Aurora - It is a display of light in the sky at the poles.

Chinook (aka snow eater) - Local warm wind of Northern America, blows on the eastern side of Rocky mountains

30. (A) Neptune

Neptune is the windiest planet in the solar system.

31. (C) Venus

Venus is also called as Lucifer, Earth's twin, morning star, evening star.

32. (B) Titan

Titan is the largest moon of Saturn.

33. (C) 27

Moon is the Earth's natural satellite. Radius - 1.74×10^6 m, distance - 3,84,000 km, gravity = $1/6$ of Earth's gravity, light from moon to Earth - 1.26 sec.

Do you know that India recently launched Chandrayaan 3 mission on Moon with Vikram lander and Pragyan rover. The point is named as Shiv Shakti point?

34. (C) Galileo Galilei

Galileo discovered **4 moons** (Io, Europa, Ganymede, Calisto) of Jupiter which are called Galilean moons.

35. (B) Pluto

Charon is the satellite of Pluto.

See solution of question 5 (level 1)

36. (D) 1.74×10^6 m

See solution of question 33 (level 1)

37. (A) Lunar exosphere

The gaseous envelope around the Moon is called Lunar Exosphere. Argon 40 is an important constituent of it.

The gaseous envelope around the Earth is called Atmosphere.

Recall the layers of Earth's atmosphere with their characteristics.

38. (D) December/January

Quadrantids are generally active between the **end of December** and the **second week of January** and **peaks** around **3-4 January**.

39. (A) April

40. (C) 90-93

International Space Station (ISS) - partner countries - Europe, USA, Russia, Canada, Japan. The largest structure put in space by humans. In 24 hours, the ISS orbits Earth 16 times. Speed - 28,000 km/hr.

41. (C) 1

The time taken by Moon for 1 rotation is same as that for 1 revolution so only one side is visible to us.

Rotation = Revolution = 27 days.

More info about moon - see solution 33

42. (A) Cygnus

See solution of question 19 (level 1)

LEVEL II

1. (D) Saturn
Saturn has 146 moon (satellites), maximum till now.
See solution of question 9 (level 1)
2. (B) Jupiter, Uranus, Neptune, Saturn
Jovian planets are composed primarily of ice, gas and liquids while terrestrial planets are composed of rocks and metals.
See solution of question 12 (level 1)
3. (D) 1, 3 and 4
Mercury doesn't have any atmosphere because of its small size, weak gravity and its proximity to the Sun. Venus doesn't have any solid surface.
Jupiter being a jovian planet is made up of gases, hence lacking any
4. (B) An equilateral triangle
Lagrange point - a point in space where the gravitational pull of two big masses equals the centripetal force of a small mass. Total 5 points - L1, L2, L3, L4, L5
L1, L2, L3 - Unstable
L4, L5 - Stable, located at same distance from both Sun and Earth.
5. (D) 32 crore kilometres
Do you know that the James Web space telescope is placed in the L2 lagrange point?
6. (C) The summer solstice occurs in northern hemisphere on June 21 and on December 21 in southern hemisphere.
Summer solstice (21 June) - rays of Sun fall vertically on Tropic of Cancer (23.5° N) - summer in NH, winter in SH
Winter solstice (22 Dec) - rays of Sun fall vertically on Tropic of Capricorn (23.5° S) - winter in NH, summer in SH
- Equinox (21 March & 23 Sept) - rays of Sun fall vertically on equator - equal day and night
7. (D) The material in the rings get condensed and form the planets that revolve in orbits around the Sun.
8. (D) 13-18 billion years
See solution of question 1 (level 1)
9. (D) Earth
Saturn is the least dense planet while Earth is the densest planet (5.51 g/cm³).
10. (A) Earth
Earth is the largest planet among the given options.
Size of planets - Jupiter > Saturn > Uranus > Neptune > Earth > Venus > Mars > Mercury.
11. (A) Copernicus
Copernicus gave Heliocentric model of universe meaning that the Earth and other planets revolve around the Sun.
Do you know that before Copernicus, Ptolemy gave Geocentric model which said that Earth is at the centre of the universe and Sun, moon revolve around it?
Laws of Kepler - (1) planets move in elliptical orbits with the Sun as a focus, (2) a planet covers the same area of space in the same amount of time, and (3) a planet's orbital period is proportional to the size of its semi-major axis.
12. (C) Earth > Venus > Jupiter > Saturn
Planets as per their density (lowest to highest) - Saturn < Uranus < Jupiter < Neptune < Mars < Venus < Mercury < Earth.
13. (C) Earth, Mercury, Mars, Venus
Terrestrial planets - Mercury, Venus, Earth, Mars
Jovian planets - Jupiter, Saturn, Uranus, Neptune

Look at the sky. We are not alone. The whole universe is friendly to us.

— Dr. A.P.J. Abdul Kalam





LATITUDE LONGITUDE ROTATION REVOLUTION



LEVEL I

1. Geoid is the _____ of the Earth. [SSC CHSL 01/07/2024]
A) Orbit B) Shape
C) Colour D) Axis
2. Which of the following statements is INCORRECT regarding the Coriolis force? [SSC CHSL 02/07/2024]
A) It deflects the wind to the right in the north and left in the south.
B) It is absent at the equator.
C) Deflection is less when the wind is high.
D) It is directly proportional to the angle of altitude.
3. Which is the Standard Meridian of India? [SSC CHSL 05/07/2024]
A) $83^{\circ} 30' E$ B) $82^{\circ} 30' E$
C) $81^{\circ} 30' E$ D) $84^{\circ} 40' E$
4. Which of the following statement/statements is/are true regarding the light year? [SSC CHSL 27/06/2024]
 1. Light year is a unit of distance.
 2. Light year is a unit of time.
 3. A light year is the distance that light travels in one Earth year.
 4. Light year is the measurement of light intensity.A) Only 1 and 3 are correct B) Only 1 and 2 are correct
C) Only 1 is correct D) Only 2 is correct
5. Perihelion is when Earth is _____. [SSC PHASE XII 24/07/2024]
A) Farthest from the sun B) Under the sun
C) Nearest to the sun D) Above the sun
6. What makes countries near the equator hotter than those which are away from the equator? [SSC JE 06/02/2024]
A) The equator is near the earth's core.
B) The equator is a hot line.
C) Areas near the equator have more volcanoes.
D) Areas near the equator receive direct and more sunlight.
7. What is the radius of the Earth? [SSC CHSL 09/03/2023]
A) 7456 km B) 6371 km
C) 5619 km D) 4728 km
8. Which among the following is not a type of solar eclipse? [SSC CHSL 15/03/2023]
A) Annular B) Partial
C) Penumbral D) Total
9. Which of the following statements regarding the moon is correct? [SSC CPO 11/11/2022]
 - I. The size of the illuminated part of the moon visible from the Earth increases each day after the new moon day.
 - II. After the full moon day, the sunlit part of the moon visible from the Earth decreases in size every day.A) Both I and II B) Only I
C) Neither I nor II D) Only II
10. During the phenomenon of aphelion, the approximate distance between the Earth and the Sun is: [SSC CHSL 12/10/2020]
A) 137 million km B) 152 million km
C) 147 million km D) 142 million km
11. How many degrees does the Earth rotate about its own axis in one hour? [SSC CGL 06/03/2020]
A) 10 B) 20
C) 24 D) 15
12. Which of the following sets of statements are correct about the Inter-Tropical Convergence Zone (ITCZ)? [SSC CPO 05/10/2023]
 - I. The ITCZ is a broad trough of low pressure in equatorial latitudes.
 - II. This is where the northeast and the southeast trade winds converge.
 - III. The ITCZ is the equatorial trough normally positioned about $5^{\circ}N$ of the equator and remains there throughout the year.
 - IV. The ITCZ moves over to the plane of Ganga in Summer.A) Only I and III B) I, II and III
C) Only II and III D) I, II and IV
13. During October-November, the apparent movement of the sun is towards the _____. [SSC MTS 18/05/2023]
A) North B) North-East
C) North-West D) South
14. On which of the following two dates does the equinox fall? [SSC CHSL 15/03/2023]
A) 21 June and 23 September
B) 21 March and 21 June
C) 21 March and 23 September
D) 23 September and 22 December
15. What is the zone between the Arctic Circle and North Pole called? [SSC CHSL 21/03/2023]
A) Torrid zone B) Frigid zone
C) North temperate zone D) South temperate zone
16. 1° latitude is equal to approximately _____ km. [SSC CHSL 16/10/2020]
A) 111 B) 145
C) 133 D) 122
17. Lines joining locations at equal travel time from a common center are called: [SSC CHSL 21/10/2020]
A) Isohalines B) Isochrones
C) Isobar D) Isobronts
18. All parallel circles from the Equator to the poles are known as _____. [SSC CHSL 26/10/2020]
A) Parallels of Latitude B) Arctic Circle
C) Tropic of Cancer D) Antarctic Circle
19. Which country has the most time zones in the world? [SSC MTS 13/10/2021]
A) France B) United States of America
C) Russia D) United Kingdom
20. Which of the following pressure belts is known as the horse latitude? [SSC CHSL 13/03/2023]
A) Equatorial Low Pressure B) Sub-Polar Low Pressure
C) High Pressure D) Subtropical High Pressure

21. On which day of the year is the earth farthest from the sun, at a position called aphelion?
 A) 19 August [SSC CHSL 19/10/2020]
 B) 23 October
 C) 12 May
 D) 4 July

22. When does the entire earth experience equal days and nights?
 [SSC CGL 04/03/2020]
 A) Days of winter solstice B) Days of summer solstice
 C) At orbital plane D) Day of equinox

LEVEL II

1. Which of the following statements is/are correct?
 1. Angular velocity for all locations on the Earth's surface is the same while linear velocity varies.
 2. Linear velocity is maximum at the equator and minimum at the poles.

Select the correct answer using the code given below.

- A) 1 only [CDS I 2024]
 B) 2 only
 C) Both 1 and 2
 D) Neither 1 nor 2

2. In order to find out the absolute location of a place on the map, which of the following will be required?
 A) Latitude of the place alone [CAPF 2021]
 B) Longitude of the place alone
 C) Both latitude and longitude of the place
 D) Neither latitude nor longitude of the place

3. Which of the following statements about maps are correct?
 1. Maps that maintain the true shapes of areas are known as conformal maps.
 2. Maps are used to show spatial relationships.
 3. Maps cannot show route from one place to another.

Select the correct answer using the code given below.

- A) 2 and 3 only [CAPF 2022]
 B) 1 and 3 only
 C) 1 and 2 only
 D) 1, 2 and 3

4. Which one of the following statements about 'great circle' is not correct?
 [CAPF 2022]
 A) Every great circle divides the Earth into equal halves.
 B) Every great circle is a circumference of the Earth.
 C) Great circles mark the longest travel routes between locations on the Earth's surface.
 D) Great circle is the largest circle that can be drawn around the Earth through two particular.

5. Which one of the following statements is correct about Spring tides?
 [CAPF 2023]
 A) High tide is higher than average high tide and Low tide is higher than average low tide.
 B) High tide is lower than average high tide and Low tide is higher than average low tide.
 C) High tide is higher than average high tide and Low tide is lower than average low tide.
 D) High tide is lower than average high tide and Low tide is lower than average low tide.

6. Which one of the following correctly explains the change in seasons on Earth?
 [CAPF 2023]
 A) Tilt of the Earth's axis
 B) Rotation of the Earth on its own axis
 C) Revolution of the Moon around the Sun
 D) Interaction of the Earth with other planets

7. Consider the following statements regarding the rotation of the Earth:
 1. Since the Earth rotates by 15° each hour, time zones normally differ by one hour.
 2. At the International Date Line, the calendar day changes - advancing a day for eastward travel, dropping back a day for westward travel.

3. Daylight saving time advances the clock by one hour.
 4. The Moon rotates and revolves about the Earth in the same direction that the Earth revolves around the Sun.

Which of the statements given above is/are correct?

[CAPF 2024]

- A) 1 only
 B) 3 and 4 only
 C) 1, 3 and 4
 D) 2, 3 and 4

8. If it is 31st December, 2023 in New Zealand, at the same time what will be the date in the Hawaiian Islands and Alaska?

[CAPF 2024]

- A) Both have 01 January, 2024
 B) Both have 30 December, 2023
 C) 31 December, 2023 in Alaska and 01 January, 2024 Hawaiian Islands
 D) 31 December, 2023 in Hawaiian Islands and 01 January, 2024 in Alaska

9. Tropic of Capricorn passes through which one of the following countries?
 [CISF AC 2022]

- A) Bolivia B) Botswana
 C) Zimbabwe D) Uruguay

10. 50° South Latitude passes through [CISF AC 2021]

- A) South Africa. B) Australia.
 C) Argentina. D) Uruguay.

11. Consider the following statements with respect to 'tides':

1. The tidal bulges on wide continental shelves have greater height.
 2. When tides are channeled between islands, they are called tidal currents.

Which of the statements given above is/are correct?

[CISF AC 2021]

- A) 1 only B) 2 only
 C) Both 1 and 2 D) Neither 1 nor 2

12. If it is 12:00 Noon in India, on which meridian will it be 7:00 am of the same day?
 [NDA I 2021]

- A) 7.5° degree E. Longitude B) 7.5° degree W. Longitude
 C) 75° degree E. Longitude D) 75° degree W. Longitude

13. Which one of the following is the longest Latitude?

[NDA I 2021]

- A) 90° degree Latitude B) 23.5° degree Latitude
 C) 0.0° degree Latitude D) 66.5° degree Latitude

14. Which one of the following is the longest parallel of latitude?
 [NDA II 2021]

- A) Tropic of Cancer B) Tropic of Capricorn
 C) Arctic Circle D) Equator

15. Which one of the following countries has the maximum time difference from Greenwich Mean Time (GMT)? [NDA II 2021]

- A) India B) Nepal
 C) Sri Lanka D) Bhutan

16. Inhabitants are unaware of the speed of rotation of the planet Earth because

1. The angular velocity is constant for each place on the Earth's surface.
 2. The atmosphere rotates with the Earth.
 3. There are no nearby objects, either stationary or moving at a rate different from that of the Earth.

Which of the above is/are the correct explanation(s)? [NDA I 2021]

- A) 1 only
B) 1 and 2 only
C) 2 and 3 only
D) 1, 2 and 3

17. Which one of the following statements with respect to Global Positioning System (GPS) is not correct? [NDA I 2021]

- A) It is based on network of satellites orbiting above the Earth.
B) It is based on the system of triangulation.
C) GPS receivers provide location in terms of latitude, longitude and altitude.
D) It provides information exclusively for military operations.

18. While watching the news, you hear about a place, of which you had never heard of before. You want to know more about the place and you want to locate it on the map. Which of the following is/are required for you to be able to locate the place on the map? [NDA I 2021]

- A) Only latitude of the place
B) Only longitude of the place
C) Both longitude and latitude of the place
D) Latitude, longitude and altitude of the place

19. In a group discussion on shape and size of the Earth, three students stated the following points:

- Student 1 : The shape of the Earth is basically an oblate spheroid.
- Student 2 : The polar diameter of the Earth is more than the equatorial diameter.
- Student 3 : Bulge along the equatorial region is due to the revolution of the Earth.

Who among the above students is/are correct? [NDA I 2022]

- A) Student 1 only
B) Student 1 and Student 2 only
C) Student 2 and Student 3 only
D) Student 1, Student 2 and Student 3

20. The periodic rise and fall of ocean water in response to gravitational forces is called [NDA II 2021]

- A) Current
B) Waves
C) Tides
D) Tsunami

21. The Prime Meridian does not pass through which one of the following African countries? [NDA II 2022]

- A) Morocco
B) Algeria
C) Mali
D) Ghana

22. Which one of the following cities of the world would represent the greatest linear velocity of rotation of the Earth? [NDA II 2022]

- A) Kampala, Uganda
B) St. Petersburg, Russia
C) Madrid, Spain
D) Stockholm, Sweden

23. What will happen if you are flying east across the International Date Line? [NDA II 2022]

- A) You will lose 12 hours
B) You will gain 12 hours
C) You will gain 24 hours
D) You will lose 24 hours

24. If it is 12 noon in New Delhi, what will be the time in London, UK? [NDA I 2023]

- A) 6 : 30 A.M.
B) 6 : 30 P.M.
C) 5 : 30 A.M.
D) 5 : 30 P.M.

25. Which of the following positions of Sun, Earth and Moon is/are suitable for Spring Tide?
1. SYZGY Conjunction
2. SYZGY Opposition
3. Quadrature

Select the correct answer using the code given below: [NDA I 2023]

- A) 1 only
B) 2 only
C) 1 and 2
D) 1 and 3

PARMAR SSC

26. Consider the following statements:

- Distance between the longitudes becomes zero on the North Pole and South Pole.
- Distance between the longitudes is maximum on the Equator.
- Number of longitudes is more than the number of latitudes.

Which of the statements given above is/are correct? [NDA I 2023]

- A) 1 only
B) 2 only
C) 1 and 3 only
D) 1, 2 and 3

27. A geographical grid has:

- Lines that run east to west around the globe to mark latitude.
- Lines that run north to south from north pole to south pole to indicate longitude.

Which of the statements given above is/are correct? [NDA II 2023]

- A) 1 only
B) 2 only
C) Both 1 and 2
D) Neither 1 nor 2

28. Consider the following statements regarding the International Date Line:

- It is roughly 180° meridian, which deviates slightly East and West to avoid land areas surrounded by the Atlantic Ocean.
- The date to the East of this line is (24 hours) earlier than to the West.

Which of the statements given above is/are correct? [NDA I 2024]

- A) 1 only
B) 2 only
C) Both 1 and 2
D) Neither 1 nor 2

29. Which of the following statements regarding neap tides is/are correct?

- It occurs every 14-15 days, which coincides with the first and third quarter of the Moon.
- This tide has a small tidal range because the gravitational forces of the Moon and the Sun are in quadrature.

Select the answer using the code given below: [NDA I 2024]

- A) 1 only
B) 2 only
C) Both 1 and 2
D) Neither 1 nor 2

30. In which among the following states in India, the Sun rays are never perpendicular?

- Bihar
- Chhattisgarh
- Manipur
- Rajasthan

Select the answer using the code given below: [NDA II 2024]

- A) 1 and 2
B) 1 and 3
C) 2 and 3
D) 2 and 4

31. Consider the following statements with respect to 'tides':
1. The tidal bulges on wide continental shelves have greater height.
2. When tides are channeled between islands, they are called tidal currents.

Which of the statements given above is/are correct? [CBSE 2023]

- A) 1 only
B) 2 only
C) Both 1 and 2
D) Neither 1 nor 2

32. Which one of the following deserts is not located around 20°-30° latitude North or South of the Equator? [GEOSCIENTIST 2022]

- A) Atacama
B) Kalahari
C) Great Victoria
D) Gobi

33. Which one of the following statements about North-facing slopes and South-facing slopes in the Northern Hemisphere is not correct? [GEOSCIENTIST 2022]
- North-facing slopes are snow covered while South-facing slopes are bare.
 - North-facing slopes get less intense radiation as compared to South-facing slopes.
 - Snow on North-facing slopes is found at a lower altitude than South-facing slopes.
 - Tree line characteristics on North facing slopes and South-facing slopes are similar.
34. You are located at 40° N latitude and 90° W longitude. You travel due North 40° , then due East 60° . What are your new coordinates? [GEOSCIENTIST 2022]
- 30° N and 80° E
 - 80° S and 30° E
 - 80° N and 30° W
 - 80° N and 30° E
35. The Prime Meridian (zero degree longitude) does not pass through which of the following country/countries?
- England
 - France
 - Portugal
 - Scotland
- Select the correct answer using the code given below: [GEOSCIENTIST 2024]
- 1 and 2
 - 2 and 3
 - 3 and 4
 - 3 only
36. In the northern hemisphere, the longest day of the year normally occurs in the: [IAS 2022]
- First half of the month of June.
 - Second half of the month of June.
 - First half of the month of July.
 - Second half of the month of July.
37. On June 21 every year, which of the following latitude(s) experience(s) a sunlight of more than 12 hours? [IAS 2024]
- Equator
 - Tropic of Cancer
 - Tropic of Capricorn
 - Arctic Circle
- Select the correct answer using the code given below:
- 1 only
 - 2 only
 - 3 and 4
 - 2 and 4
38. If it is 10:00 a.m. at Greenwich Mean Time (GMT), then what will be the time at 90° E longitude? [CDS I 2023]
- 5 : 00 a.m.
 - 4 : 00 a.m.
 - 3 : 00 p.m.
 - 4 : 00 p.m.
39. Consider the following lines of Longitude and Latitude:
- Prime Meridian
 - Tropic of Cancer
 - Equator
- Which of the above lines is/are a Great Circle? [NDA I 2025]
- 1, 2 and 3
 - 1 and 2 only
 - 1 and 3 only
 - 3 only
40. Which one of the following latitudes will experience a minimum angle of the Sun's rays when it is Summer Solstice in the Northern Hemisphere? [NDA I 2025]
- Arctic Circle
 - Equator
 - Tropic of Cancer
 - Tropic of Capricorn

@ujjaval012

SOLUTIONS

LEVEL I

1. (B) Shape

The shape of Earth is **Geoid/Oblate** spheroid (flat from top and bottom and bulges at centre)

Reason for geoid shape - more centrifugal force at equator due to rotation.

2 types of forces act on a revolving body - Centripetal force (towards the centre) and Centrifugal force (away from centre)

2. (C) Deflection is less when the wind is high

Coriolis force is the apparent force experienced due to rotation of Earth. It deflects the wind to the right in the NH and to the left in the SH. It is **max** at **poles** and **absent** at equator. It is directly **proportional** to the speed of wind.

Geostrophic winds - winds which blow parallel to isobars due to balance of Coriolis force and pressure gradient

3. (B) 82°30'E

Meridian (longitude) is the imaginary **vertical** lines running from North to South. It is the angular distance of a place from prime meridian. **Distance** between meridians - **max** at equator (111.32 km) and **least** at poles. Total meridians = 360.

Important meridians - Prime meridian/GMT (0°) - passes through Greenwich, London ; International Date Line (180°)

GMT passes through 8 countries - UK, France, Spain, Algeria, Mali, Burkina Faso, Togo, Ghana (TRICK - BSF GAME in TOGO Kingdom). Moving from W to E on IDL - day gained and date lost.

Do you know that all **longitudes** are **great circles** (largest circle which can be drawn on a sphere dividing it into two equal halves) but only **1 latitude** (equator) is a **great circle**?

Tropic of Cancer and Indian Standard Meridian intersect each other at Koriya district in Chattisgarh.

4. (A) Only 1 and 3 are correct

Light year (9.46×10^{12} years) is the distance travelled by light in 1 Earth year. Hence, it is a unit of distance.

1 Parsec = 3.26 light years.

5. (C) Nearest to the Sun

Perihelion (3 January) - Earth is nearest to the Sun (147.5 mn km)

Aphelion (4 July) - Earth is farthest from the Sun (152.5 mn km)

Distance between Sun and Earth (avg) - 150 mn km

Perigee - moon closest to Earth **Apogee** - moon farthest from Earth

6. (D) Areas near the equator receive direct and more sunlight.

Equator (0° latitude) divides the Earth in two equal halves - NH and SH.

Latitudes - imaginary horizontal lines running East to West. It is the angular distance of a place from equator.

1° latitude = 111 km. Total latitudes = 181. Distance between each latitude is same.

Important latitudes - 0° - Equator, 23.5° N - Tropic of Cancer, 23.5° S - Tropic of Capricorn, 66.5° N - Arctic circle, 66.5° S - Antarctic circle

Do you know that the direct sunlight only falls between 23.5° N and 23.5° S. Beyond these, only slant rays fall?

The temperature in subtropical regions is more than that at equator due to factors like less cloudiness in subtropics and absence of large water bodies.

7. (B) 6371 km

Equatorial radius = 6378 km	Equatorial circumference = 40,075 km
Polar radius = 6357 km	Polar circumference = 40,007 km
Mean radius = 6371 km	Mean circumference = 40,040 km

Equatorial speed > Polar speed, Equatorial radii > Polar radii, Equatorial circumference > Polar circumference

8. (C) Penumbral

Solar Eclipse - moon is between **Sun and Earth** and blocks the sunlight. Occurs during New moon (amavasya).

It is of 3 types - total, annular, partial

Lunar Eclipse - earth is between Sun and moon and blocks the sunlight. Occurs during Full moon (purnima)

Blue moon is when there are 2 purnima in one month

Lunar eclipse + perigee = bigger moon (super moon)

9. (A) Both I and II

Full moon (purnima) - moon appears as a full circle

New moon (amavasya) - moon is invisible

After purnima, illuminated part of moon decreases (waning phase) each day and in nearly **15 days** it becomes invisible thereby causing amavasya.

10. (B) 152 mn km

Perihelion (3 January) - Earth is **nearest** to the Sun (147.5 mn km)

Aphelion (4 July) - Earth is **farthest** from the Sun (152.5 mn km)

11. (D) 15

Each hour, Earth rotates about 15° and hence in 24 hours it completes one rotation ($24 \times 15^\circ = 360^\circ$).

Moving 15° East from prime meridian - time increases by 1 hour

Moving 15° West from prime meridian - time decreases by 1 hour

Rotation causes day and night while revolution and tilt cause change in seasons.

12. (D) I, II and IV

ITCZ (doldrum) is a **low pressure** belt between 10° N and 10° S. It is a zone of **convergence** of airmass. It moves north and south with the apparent shift of Sun.

Cell - representation of general circulation of atmosphere. 3 cells - Hadley cell, Ferrel cell, Polar cell

13. (D) South

During October-November, the apparent movement of the Sun is towards the South because during this time, Earth is tilted away from the Sun in NH and towards the Sun in SH.

Earth is tilted at **23.5°** on its axis and **66.5°** on its orbital plane.

Summer solstice - 21 June, Winter solstice - 22 December

14. (C) 21 March and 23 September

Equinox - sunlight fall vertically on equator, so equal day and night. Vernal/Spring equinox-21 march, Autumnal/fall equinox - 23 september. Recall about summer and winter solstice.

15. (B) Frigid zone

Frigid zone - 66.5° N - 90° N and 66.5° S - 90° S

Torrid zone - 23.5° N - 23.5° S

Temperate zone - 23.5° N - 66.5° N and 23.5° S - 66.5° S

Recall all the important latitudes.

16. (A) 111 km

$1^\circ \text{ latitude} = 111 \text{ km}$

17. (B) Isochrones

Isochrones - line joining locations at equal travel time

Isohalines - line joining places of same salinity

Isobar - line joining places of same atmospheric pressure

Isobronts - line joining places where thunderstorms are in a specific phase at a specific time

Isohyets - line joining places with same amount of rainfall

Isonephs - line joining places with same amount of cloud cover

Isobaths - line joining places with same depth below water

18. (A) Parallels of Latitude

See solution 6

Arctic circle - 66.5° N , **Antarctic circle** - 66.5° S , **Tropic of Cancer** - 23.5° N , **Tropic of Capricorn** - 23.5° S

Do you know that longitudes intersect each other at North and South poles whereas latitudes don't intersect each other?

19. (A) France

France has 13 time zones due to its overseas territories.

Russia - 11 USA - 6 UK - 1

India has only 1 time zone i.e. $82^\circ 30' \text{ E}$

20. (D) Subtropical High Pressure

Pressure belts are of 4 types - Equatorial low pressure belt, Subtropical high pressure belt (30° N and 30° S), Subpolar low pressure belt (60° N and 60° S) and Polar high pressure belt (80° N and 80° S)

Do you know that major hot deserts in the NH are located between 20° - 30° N because of subtropical high pressure (the subsiding air is warm and dry)?

21. (D) 4 July

Perihelion (3 January) - Earth is nearest to the Sun (147.5 mn km)

Aphelion (4 July) - Earth is farthest from the Sun (152.5 mn km)

22. (D) Day of equinox

Equinox - sunlight fall vertically on equator, so equal day and night. Vernal/Spring equinox - 21 march, Autumnal/fall equinox - 23 september

LEVEL II

1. (C) Both 1 and 2

Angular velocity - how fast an object moves relative to a point (in Earth's case relative to its axis). It is same (24 hours) for all locations on Earth.

Linear velocity - rate at which an object moves along its path. It is max at equator and min at poles (due to bulge at the equator)

2. (C) Both latitude and longitude of the place

We can easily find the absolute location of a place by the intersection point of latitude and longitude. So, both are required.

Recall all info about latitude and longitude.

3. (C) 1 and 2 only

Conformal maps - these maps keep the shape same but may change the size. (Imagine drawing a small square on a balloon. If you blow up the balloon, the square gets bigger, but its corners still make 90° angles.) Maps can show route from one place to another. Such maps are called road maps.

4. (C) Great circles mark the longest travel routes between locations on the Earth's surface.

Great circle is the largest circle which can be drawn on. It divides the Earth in two equal halves. It marks the shortest distance between two points along the sphere.

Do you know that equator is the only latitude which is a great circle whereas all longitudes are great circles?

5. (C) High tide is higher than average high tide and low tide is lower than average low tide.

Spring tide - creates the greatest difference in water levels between high and low tides.

Neap tide - creates the smallest difference in water levels between high and low tides. Occurs approx 2 times a month (during the first and third quarter moon phase)

6. (A) Tilt of the Earth's axis

Change in season on Earth is caused by tilt of Earth's axis and revolution around the Sun while day and night are caused due to rotation of Earth on its axis.

Recall the axial and orbital tilt of Earth.

Which planet is called the rolling planet due to its tilt of 98° ?

7. (C) 1, 3 and 4

Earth rotates 360° in 24 hours. So, 1 hour = 15° . Hence, time zones normally differ by one hour.

At the IDL, day is dropped back for eastward travel and advanced for westward travel.

Daylight saving time - it is the practice of forwarding clock by 1 hour in summers and back again by 1 hour in fall to make the better use of daylight.

The moon also rotates and revolves around the Earth in the same direction that the Earth revolves around the Sun (Earth rotates from W to E)

8. (B) Both have 30 December, 2023

New Zealand is located just west of IDL while Hawaiian Islands and Alaska are located to the east of IDL. On moving from west to east on IDL, day is dropped back.

9. (B) Botswana

Tropic of Capricorn (23.5° S) passes through 10 countries.

South America : Argentina, Brazil, Chile, Paraguay

Africa : Namibia, Botswana, South Africa, Mozambique, Madagascar

Australia : Western Australia, Northern Territory, Queensland

Do you know that Africa is the only continent through which 23.5° N , 0° , 23.5° S all three pass?

10. (C) Argentina

50° S latitude passes through Argentina.

11. (C) Both 1 and 2

The tidal bulge on wide continent shelves have greater height because water depth decreases abruptly. Hence, height increases to conserve the energy.

When the tide is channeled between islands or into bays and estuaries they are called tidal currents.

Tides are the periodic rise and fall of sea level, once or twice a day, mainly due to the attraction of the Sun and the moon.

12. (A) 7.5° E Longitude

Since $15^\circ = 1$ hour, so 5 hours ($12-7 = 5$ hours) = 75° .

IST is 82.5° E so to go back 5 hours = 82.5° E - $75^\circ = 7.5^\circ$ E

13. (C) 0° Latitude

Equator (0°) is the longest latitude while poles (90°) are the smallest latitudes.

14. (D) Equator

Equator (0°) is the longest latitude while poles (90°) are the smallest latitudes.

15. (D) Bhutan

Time difference from GMT increases as we move away from it. Bhutan is farthest from it among the options.

Moving West from GMT - time decreases

Moving East from GMT - time increases

16. (D) 1, 2 and 3

Angular velocity is same for all the places on Earth.

All objects on Earth and atmosphere moves with Earth, so we are unaware of Earth's rotation.

1 rotation = 23 hours 56 mins 4 sec; Rotational speed is max at equator and min at poles.

17. (D) It provides information exclusively for military operations.

GPS is a system of interconnected satellites used to provide location. It was developed for military operations but now is freely available for civilian use.

A system of triangulation is a surveying method where a network of interconnected triangles is established to determine the location of points by measuring angles between them.

NavIC (Navigation with Indian Constellation) - it is Indian navigation system similar to GPS.

18. (C) Both longitude and latitude of the place

We can easily find the absolute location of a place by the intersection point of latitude and longitude. So, both are required.

19. (A) Student 1 only

See solution of questions 1 and 7 (level 1)

20. (C) Tides

See solution 11

Tides based on frequency - Semi diurnal tide - 2 high and 2 low tides each day; Diurnal tide - 1 high and 1 low tide each day; Mixed tide.

Tides based on height - Spring tide - greatest difference between high and low tide; Neap tide - smallest difference between high and low tide.

Tsunamis are giant waves caused by earthquakes or volcanic eruptions under the sea.

21. (A) Morocco

See solution of question 3 (level 1)

22. (A) Kampala, Uganda

Linear velocity is higher near equator due to larger radius at equator. Kampala is located near equator so it would represent the greatest linear velocity among the given options.

Recall about angular velocity.

23. (C) You will gain 24 hours

When we fly east across the International Date Line, we gain 24 hours.

24. (A) 6:30 A.M.

Difference between New Delhi and London is 82.5° which translates to 5.5 hours. Since London is to the west of New Delhi so time there will be behind. So $12 - 5.5 = 6:30$ A.M.

25. (C) 1 and 2

Spring tide occurs when Sun, Earth and Moon are in straight line (SYZYG condition).

Syzygy conjunction - Moon between Sun and Earth

Syzygy opposition - Earth between Sun and moon

Quadrature means when Sun, Earth and moon are at 90° . It is associated with Neap tides.

26. (D) 1, 2 and 3

See solution of questions 3 and 6 (level 1)

27. (C) Both 1 and 2

Geographical grid is formed when latitudes and longitudes intersect each other forming a grid. It is used to locate a place on the map.

Recall about latitudes and longitudes.

28. (B) 2 only

See solution 3

Statement 1 is partially correct. IDL moves East or West to avoid land areas surrounded by Pacific ocean and not the Atlantic ocean.

29. (C) Both 1 and 2

See solution of questions 5 and 18 (level 2)

30. (B) 1 and 3

Sun rays are perpendicular between 23.5° N (Tropic of Cancer) and 23.5° S (Tropic of Capricorn). Bihar and Manipur lie above Tropic of Cancer, so they always experience slant rays.

Tropic of Cancer passes through 8 Indian states - Gujarat, Rajasthan, MP, Chattisgarh, Jharkhand, West Bengal, Tripura and Mizoram.

Do you know that Mahi is the only Indian river to cross Tropic of Cancer twice?

31. (C) Both 1 and 2

See solution of question 11 (level 2)

32. (D) Gobi

Gobi desert is not a tropical desert. It is located on the border of China and Mongolia.

Locations of other deserts - Atacama desert - Northern Chile. Kalahari - South Africa, Great Victoria - Australia

3 major deserts of Australia - Sandy, Gibson, Victoria (TRICK - 7GB)

33. (D) Tree line characteristics on North facing slopes and South facing slopes are similar.

In NH, the northern slope of mountains receive less sunlight as compared to the southern slope, hence northern slope is snow covered. North facing slopes have lower tree line as compared to south facing slopes (tree line is the line on a mountain above which the trees don't grow).

34. (C) 80°N and 30°W

Current location - 90°W. When we move 60° east then we are moving towards the prime meridian so now our current position will be 90°W - 60°E = 30°W

35. (C) 3 and 4

The prime meridian passes through UK, France, Spain, Algeria, Mali, Burkina Faso, Togo, Ghana.

36. (B) Second half of the month of June

In the NH, the longest day is on 21 June (summer solstice) while in the SH, the longest day is on 22 December (winter solstice). On equinox, day and night are equal.

37. (D) 2 and 4

On 21 June (summer solstice), Sun is vertically over Tropic of Cancer (23.5°N). Hence NH experiences the **longest day** which lasts over 12 hours. During this time, **North Pole** experiences **daylight of 24 hours**, a phenomena known as 'Midnight sun'.

Equator experiences the daylight of nearly 12 hours regardless of the position of the Sun.

Sun is vertically over Tropic of Capricorn (23.5°S) on 22 December (winter solstice).

38. (D) 4:00 p.m.

Since 15° = 1 hour so 90° = 6 hours. Adding 6 hours to 10:00 a.m. we get 4:00 p.m.

@ujjaval012

Lines on a map may seem invisible, but they hold the world together in order and understanding.

— Anonymous